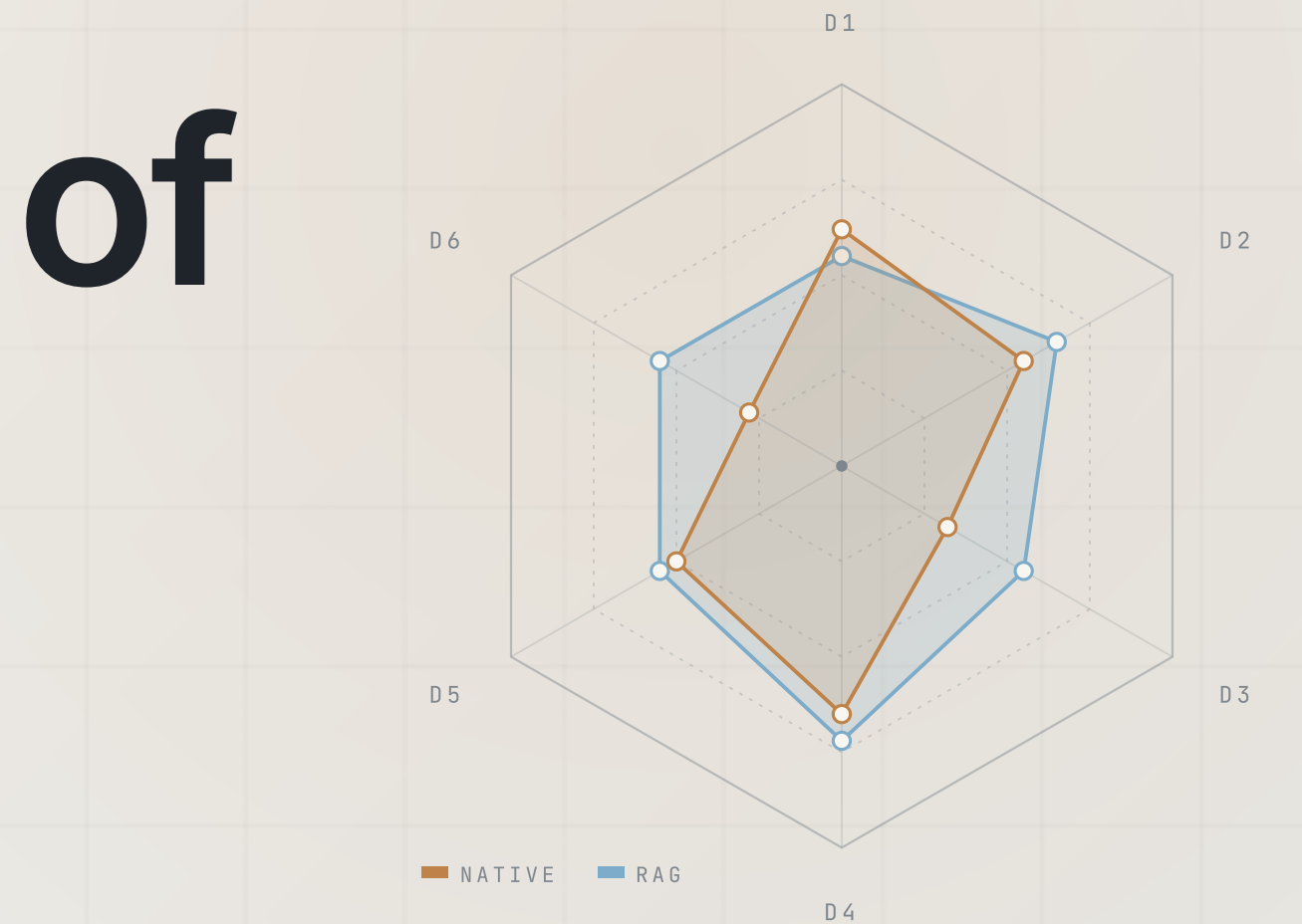


— TERM 2 PRESENTATION

Domain Adaptation of *Llama-Primus* on Singapore's Cybersecurity Code of Practice *(CCoP 2.0)*



— CONTENTS

Today's *agenda*.

03	Recap	—	04	Roadmap	—
05	Architecture diagram	—	06	Models & tool choices	—
07	Ground truth quality	—	08	Document ingestion pipeline	—
09	Retrieval pipeline	—	10	RAG augmentations	—
11	RAGAs framework	—	12	LLM as Judge	—
13	Evaluation results — Native vs RAG	—	14	Per-benchmark scores	—
15	Worked example	—	16	Observations	—
17	Next steps	—	18	Term 3 preview · RAFT	—

RECAP

From *fine-tune first* to RAG-first.

TERM 1 · SEPT - DEC 2025

Established the case for fine-tuning Llama-Primus-Reasoning on CCoP 2.0.

01 Foundation & setup

Local M3 evaluation infra; quantised Primus-Reasoning (8B, 4-bit) running offline.

02 CCoP 2.0 mapped

220 auditable clauses across 11 sections; IT and OT scope; ground-truth set drafted.

03 Baseline benchmarked

48.96% weighted across 17 of 21 benchmarks — reasoning 61.6%, factual grounding 31.6%.

04 Failure modes named

Hallucination of regulatory facts, scope confusion, ungrounded paraphrase.

OPEN QUESTION CARRIED FORWARD

Reasoning works; factual grounding does not. Fine-tuning encodes regulation implicitly — it cannot supply the authoritative clause text the audit requires.

PIVOT

Lead with RAG.
Quantify the lift.
Fine-tune the
residual gap.

STRATEGIC REORDER ·
JANUARY 2026

TERM 2 · JAN - APR 2026

Reordered the program: prove RAG's contribution, then fine-tune what RAG can't fix.

01 Build RAG over CCoP 2.0

Local Qdrant hybrid search, clause-aware chunking of 8 authoritative documents, cross-encoder rerank, diagram captioning.

02 Upgrade evaluation

Dual-layer scoring — RAGAs automated quality metrics plus LLM-as-Judge for reasoning depth and hallucination.

03 Quantify native vs RAG

Same model, same ground-truth set; measure how much factual grounding retrieval recovers across 21 benchmarks.

04 Target fine-tuning to the residual

Benchmarks where RAG-augmented responses still fall short become the focused fine-tuning dataset.

WHY THIS SHAPE

Retrieval supplies regulatory grounding directly. Fine-tuning is reserved for what only fine-tuning can fix — making the eventual training budget defensible.

ROADMAP

Project timeline across *three phases*.



PROJECT TIMELINE • SEPT 2025 → AUG 2026

● YOU ARE HERE • APRIL 2026



ARCHITECTURE

LangGraph pipeline enabling *dual inference modes*.

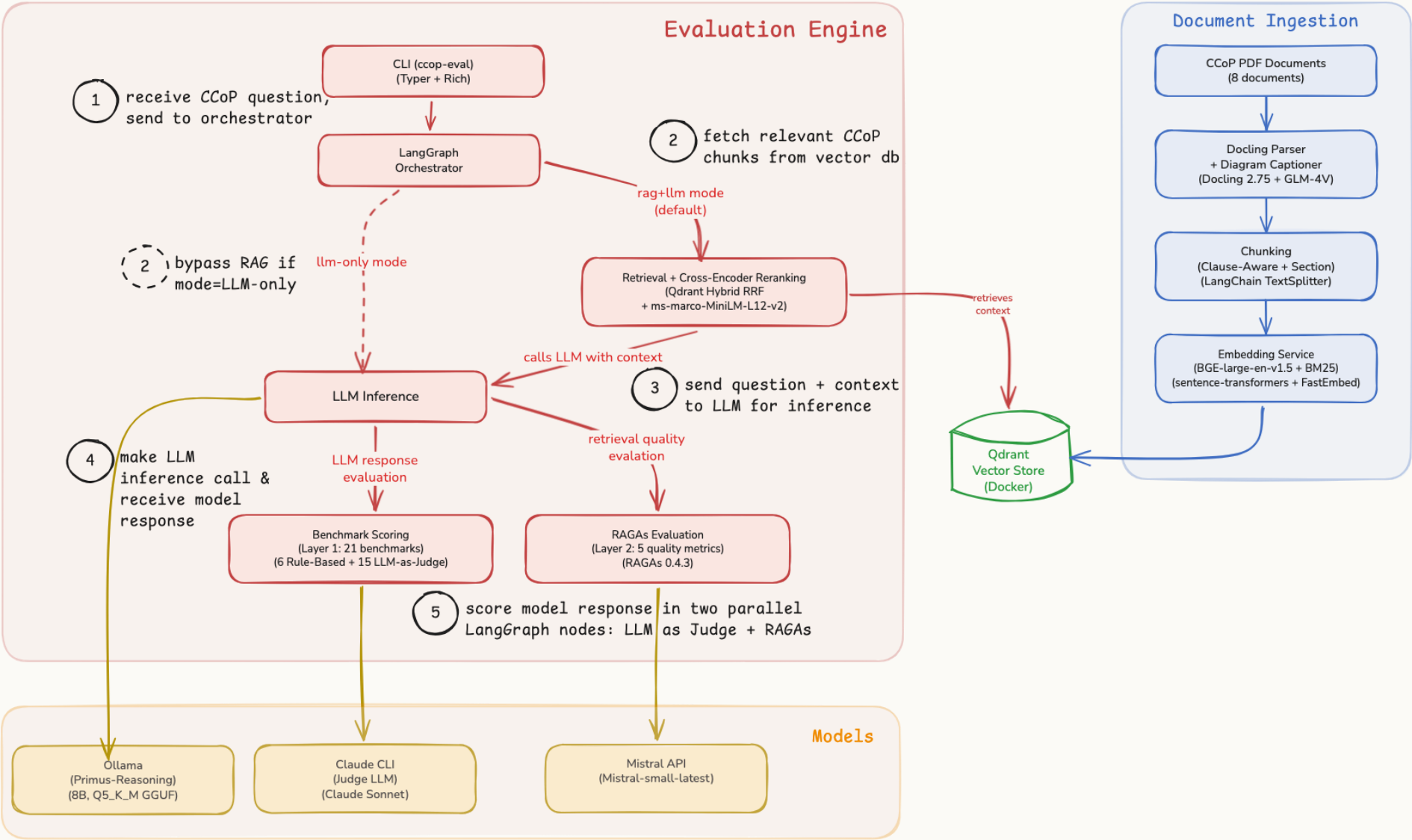


FIGURE 1 · §11 APPLICATION ARCHITECTURE

THREE SUBSYSTEMS

A Evaluation Engine

- CLI · LangGraph · retrieval + reranking
- LLM inference · two-layer scoring
- rag+llm and llm-only modes

B Document Ingestion

- Docling parser + GLM-4V captioning
- Clause-aware chunking
- BGE-large + BM25 dual-encode

C Vector store

- Qdrant · Docker · local
- 495 chunks · seven-document corpus
- RRF-compatible hybrid index

D External models

- Ollama · Primus-Reasoning 8B (under test)
- Qwen3-235B-A22B (Judge, OpenRouter)
- Mistral Small (RAGAs evaluator)

MODELS & TOOLS

Models, Tools & *Runtime Framework*.

REQUEST-PATH MODELS · INDEPENDENT & NON-OVERLAPPING

UNDER EVALUATION	Llama-Primus-Reasoning · 8B Q5_K_M GGUF · served locally via Ollama	OLLAMA · LOCAL
LLM AS JUDGE	Qwen3-235B-A22B qwen/qwen3-235b-a22b-07-25 · temp 0.2 · different model family from system under test	OPENROUTER
RAGAS EVALUATOR	Mistral Small mistral-small-latest · scores response & retrieval quality metrics	MISTRAL API

INDEXING-TIME MODELS · RUN ONCE

CONTEXTUAL RAG · CHUNK CONTEXT	GPT-4o-mini Indexing-time · 2–3 sentence summary block prepended to every chunk before embedding (Anthropic, 2024)	OPENROUTER
HYDE · QUERY REWRITE	GPT-4o-mini Request-time · rewrites the user query into a hypothetical clause-style answer for embedding	OPENROUTER
DIAGRAM CAPTIONING	GLM-4V 78 diagrams across the corpus · indexing-time only · graceful fallback on API failure	GLM API

LOCAL STACK

VECTOR DB Qdrant · Docker · local	DENSE EMBED BAAI/bge-large-en-v1.5 · 1024d	SPARSE EMBED Qdrant/BM25 · FastEmbed
RERANKER bge-reranker-large · cross-encoder	ORCHESTRATOR LangGraph · stateful pipeline	CLI Typer + Rich · poetry run ccop-eval

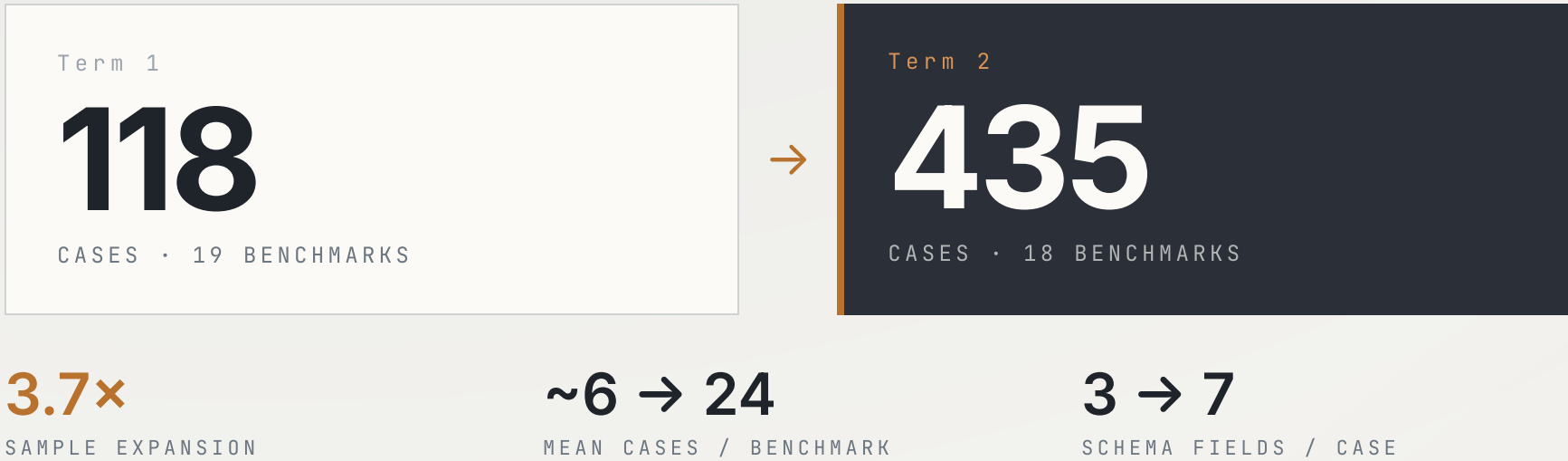
OPERATING THE FRAMEWORK

SETUP <pre>docker compose up -d qdrant ollama pull primus-reasoning poetry run ccop-eval setup check</pre>
THREE RUN MODES HYBRID DEFAULT <pre>poetry run ccop-eval evaluate run --mode hybrid</pre> HyDE → retrieval → rerank → generation → dual scoring <i>Default · headline benchmark of the RAG pipeline</i>
LLM-ONLY <pre>poetry run ccop-eval evaluate run --mode llm-only</pre> Bypass retrieval · model answers from parametric memory <i>Baseline for measuring RAG contribution (§14.3 ablation)</i>
RAG-ONLY <pre>poetry run ccop-eval query ask --mode rag-only</pre> Retrieval without LLM · returns top-3 grounded passages <i>Ad-hoc clause lookup · retrieval debugging</i>

GROUND TRUTH

Expansion of *ground truth* & schema for evaluation.

TERM 1 → TERM 2 • CARDINALITY



SCHEMA FIELDS • CALIBRATING THE JUDGE

FIELD	DIM	ROLE
expected_label	D1	Verdict — compliant / non-compliant / not-applicable
key_facts (tiered)	D1 • D2	Critical / important / supporting — missing CRITICAL costs more
reasoning_chain	D2	Step-by-step trace of expected reasoning
forbidden_claims	D3	Auto-CONTRADICTED on substring match
hallucination_patterns	D3	Regex anchors for deterministic hallucination detection
clause_reference	D6	Canonical clause IDs validated against the inventory

THREE DRIVERS

- 01

Benchmark consolidation
Code-quality, safety, training, performance families dropped — out of regulatory-reasoning scope.
- 02

Per-benchmark expansion
Mean per-benchmark cardinality up from ~6 to 24 — makes benchmark-level deltas statistically interpretable.
- 03

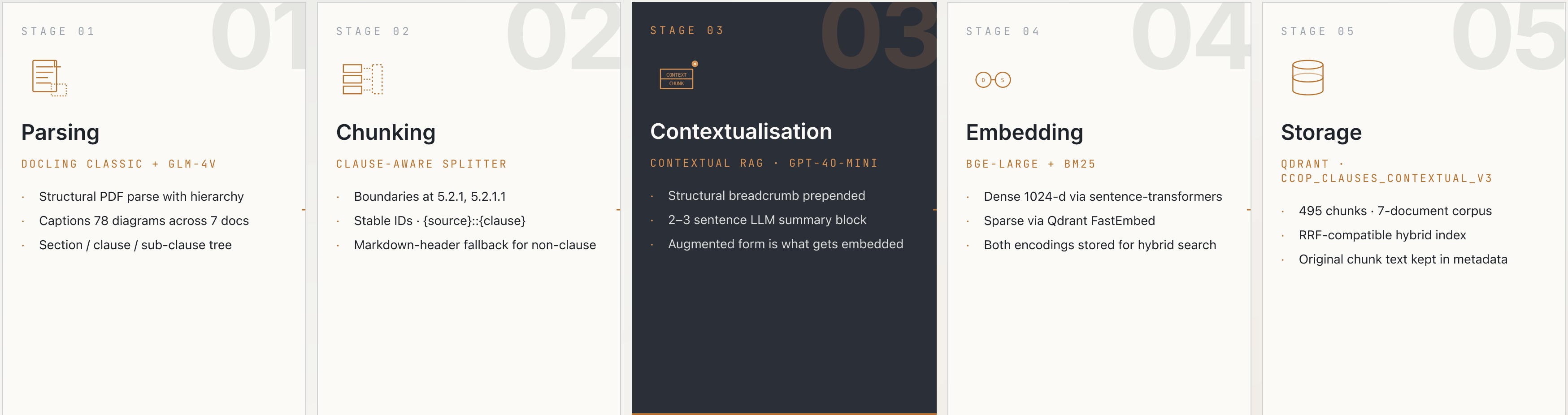
Schema enrichment
Six new fields, each calibrated to a judge dimension — flat triple → anchored multi-dim rubric.

VALIDATION • 3 LAYERS

Schema check • auditor + verifier pair • stratified n=18 (bdc4927d)

— RAG INDEXING PIPELINE

Document ingestion pipeline — *indexed once.*



CORPUS

7 documents · 495 chunks

CCoP 2.0
PRIMARY REGULATORY TEXT

CCoP RTF
RESPONSE TO FEEDBACK

Cybersecurity Act 2018
STATUTORY GROUNDING

Auditing Guidelines
AUDIT REFERENCE

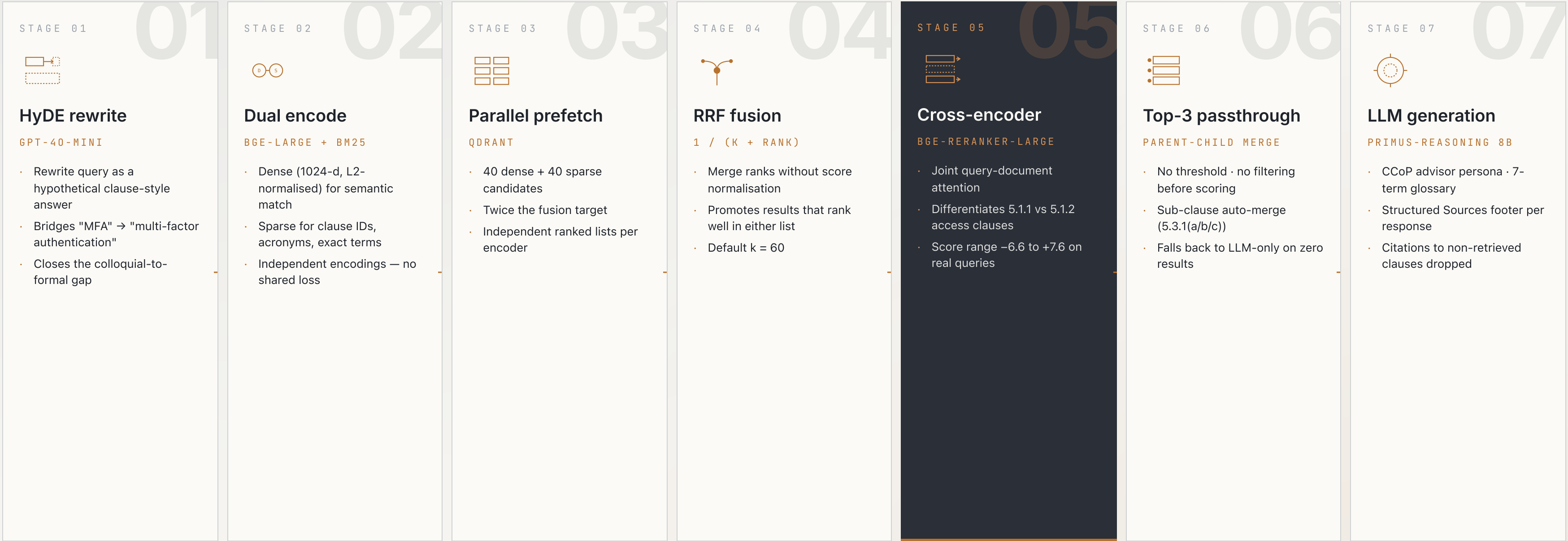
Threat Modelling Guide
RISK PRACTICE

Risk Assessment Guide
RISK PRACTICE

Security By Design
ENGINEERING REFERENCE

RETRIEVAL PIPELINE

Seven-stage pipeline run on *every query*.



LOCAL STACK

495 chunks · 7 documents

VECTOR DB
Qdrant

DENSE
BGE-large-en-v1.5

SPARSE
BM25 · FastEmbed

ORCHESTRATOR
LangGraph

INFERENCE
Ollama · Llama-Primus 8B

INDEXING
Docling Classic + GLM-4V

— TWO AUGMENTATIONS

Augment *chunks* at index, *queries* at request.

A • INDEXING TIME

once

Contextual RAG

Each chunk is augmented with a structural breadcrumb plus a 2–3 sentence LLM summary *before* embedding. Anchors the embedding in its parent context — the cross-encoder can now discriminate semantically close clauses on short queries.

RAW CHUNK (AS PARSED)

5.2.2 The CII0 shall perform a periodic review of all accounts and revoke access privileges that are no longer required.

↓ GPT-40-MINI PREPENDS BREADCRUMB + CONTEXT

AUGMENTED CHUNK (WHAT GETS EMBEDDED)

[CCoP 2.0 → Chapter 5: Identity and Access Management → 5.2.2]
This clause obligates the CII0 to run periodic account reviews and revoke stale privileges. Search terms users might use: "account review", "stale access", "privilege revocation".

5.2.2 The CII0 shall perform a periodic review of all accounts and revoke access privileges...

EFFECT "5.2.2" no longer ambiguous between unrelated documents · embedding sees regulatory framing

B • QUERY TIME

per request

HyDE rewrite

The user's question is rewritten as a hypothetical clause-style answer; that hypothetical is embedded for retrieval. Bridges acronym expansion and colloquial-to-formal phrasing gaps that bi-encoders cannot perform via attention.

USER QUERY (COLLOQUIAL)

"What are the MFA requirements?"

↓ GPT-40-MINI REWRITES IN FORMAL CCOP STYLE

HYPOTHETICAL CLAUSE (WHAT GETS EMBEDDED)

The CII0 shall implement multi-factor authentication for all privileged accounts accessing critical information infrastructure systems. The Commissioner may require additional authentication factors where the residual risk warrants heightened controls under Section 11(7).

EFFECT "MFA" → "multi-factor authentication" · embedding-space comparison is regulatory-style on both sides

RAGAS

Three questions about *retrieval*.

EVALUATOR

Mistral Small

RAGAs framework · RAG-mode only

SCOPE

Retrieval only

Did the right context surface and get used?

SCALE

0.0 – 1.0

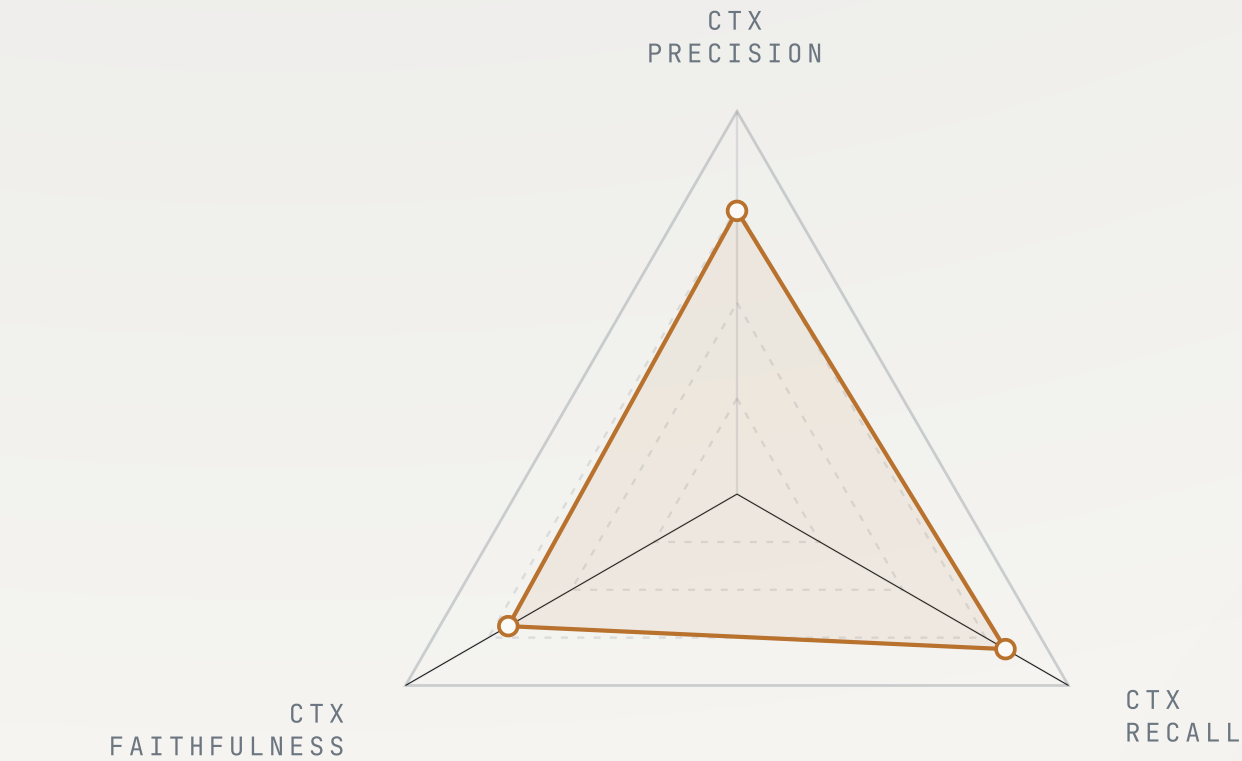
Per-question continuous scores, then averaged

COVERAGE

RAG-mode runs only

Per query alongside Judge dimensions; ablation on the n=18 stratified sample

SAMPLE RUN · RAGAS SCORES



RINGS: 0.25 → 0.50 → 0.75 → 1.00

THREE RETRIEVAL-QUALITY METRICS

01 context_precision

Context precision

"Are retrieved chunks relevant?"

FAILURE MODE Low → noise in the top-N

02 context_recall

Context recall

"Do retrieved chunks contain what is needed?"

FAILURE MODE Low → the right chunks were missed

03 context_faithfulness

Context faithfulness

"Does the response use what was retrieved?"

FAILURE MODE Low → model is ignoring the context

LLM AS JUDGE

Six dimensions, *one rubric*, anchored to ground truth.

JUDGE MODEL

Qwen3-235B-A22B

OpenRouter · temp 0.2

SCALE

0–3 anchored

Per-dimension anchors in Appendix C

APPROACH

Single rubric

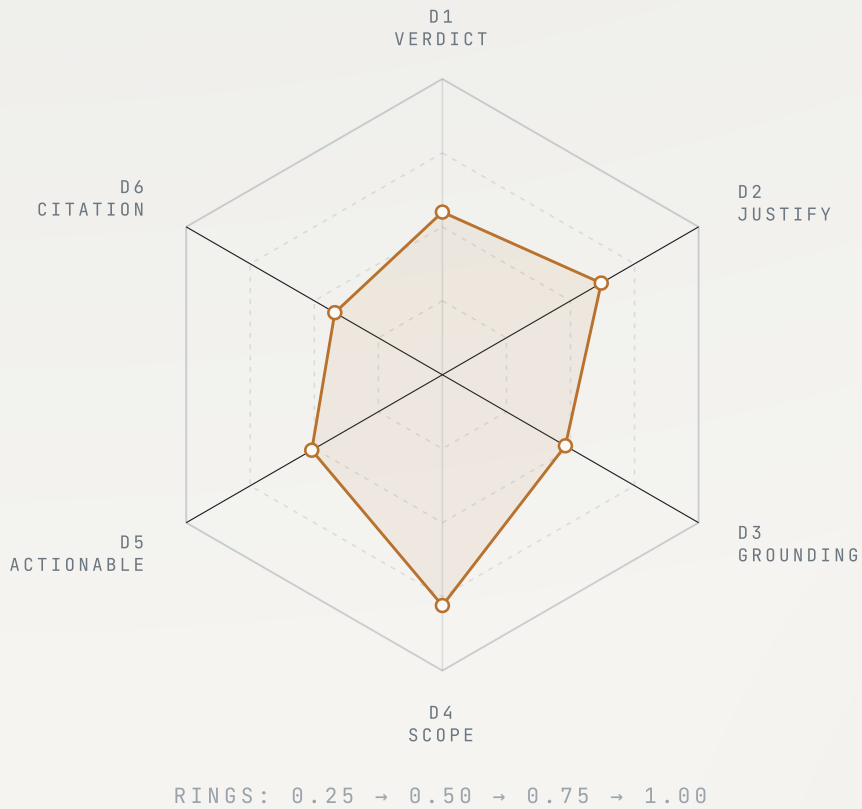
Benchmark-agnostic; signal from GT payload

INDEPENDENCE

Different family

No self-evaluation; secondary judge: gpt-4o-mini

SAMPLE RUN · PER-DIMENSION SCORES



SIX DIMENSIONS · D3 & D6 ARE COUNT-BASED

D1 verdict_accuracy

"Does the final verdict match the expected answer?"

ANCHOR expected_label · key_facts

D2 justification_quality

"Is the reasoning logically sound and internally consistent?"

ANCHOR reasoning_chain · key_facts

D3 factual_grounding

"Are atomic claims supported by context, key_facts, or cited clause text?"

ANCHOR forbidden_claims · hallucination_patterns

D4 scope_appropriateness

"Does the response stay within scenario constraints?"

ANCHOR scenario_sector · scenario_role

D5 actionable_way_forward

"Is the analysis translated into specific next steps?"

ANCHOR expected_response

D6 citation_correctness

"Are clause IDs real and accurately described?"

ANCHOR clause_reference · expected_citations

"Real clauses, wrong descriptions"
fails D6 but passes D3 — that's the failure mode the rubric makes legible.

EVALUATION RESULTS

Six-dimension *Native vs RAG* comparison.

SETUP

Ran a stratified **18-case sample** covering each active benchmark · same model (Llama-Primus), same judge (Qwen3-235B), same six-dimension rubric.

NATIVE

0.392

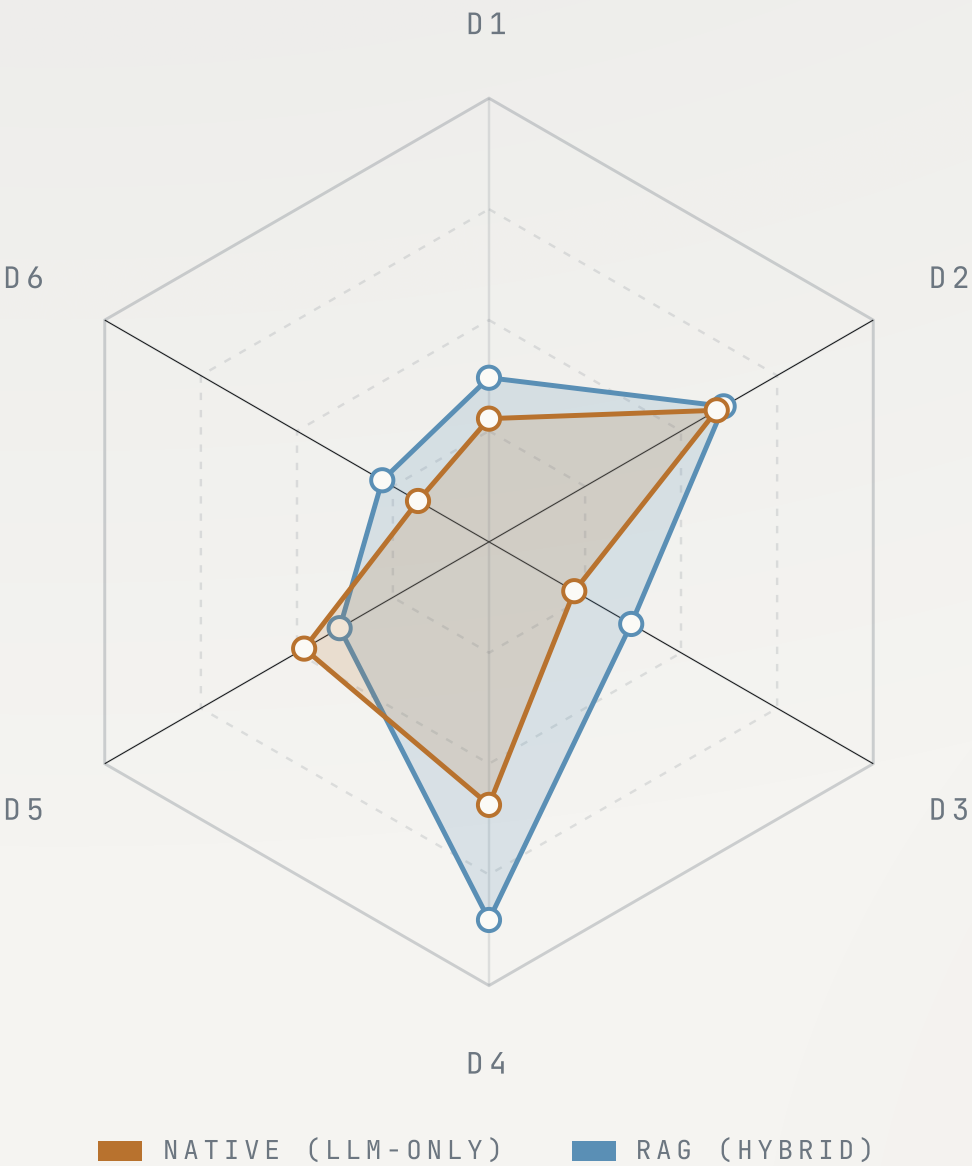
RAG

0.478

DELTA Δ

+0.086

PER-DIMENSION MEANS · RADAR



D1 verdict_accuracy NATIVE 0.278 RAG 0.370 +0.092 Modest lift; both modes weak — verdict still leans on parametric memory.	D2 justification_quality NATIVE 0.593 RAG 0.611 +0.018 Reasoning is parametric — barely moves with retrieval.
D3 factual_grounding NATIVE 0.222 RAG 0.370 +0.148 Clause grounding lifts atomic-claim correctness.	D4 scope_appropriateness NATIVE 0.593 RAG 0.852 +0.259 Largest gain — retrieval anchors response to the question.
D5 actionable_way_forward NATIVE 0.481 RAG 0.389 -0.092 Slight regression — RAG more conservative on prescriptive next-steps.	D6 citation_correctness NATIVE 0.185 RAG 0.278 +0.093 Improved but still low — misattribution persists.

EVALUATION RESULTS

Per-benchmark scores • *Native vs RAG.*

MEAN Δ		STRONG WINS		FLAT / MARGINAL		RAG-NEGATIVE	
+0.086		4		9		5	
Headline RAG lift		Δ ≥ +0.30 — B18, B13, B02, B21		Δ within ±0.10		B23, B04 lead the regression	
B - ID	DOMAIN	NATIVE		RAG		Δ	FOUR INSIGHTS
B18	Responsibility attribution (SG)		0.11		0.89	+0.78	01 Strong help on regulatory specifics Four benchmarks > +0.30. B18 (governance roles +0.78), B13 (board oversight evidence +0.44), B02 (compliance classification +0.33), B21 (hallucination resistance +0.33) — all need precise CCoP text the model cannot recall from training.
B13	Evidence expectation awareness		0.17		0.61	+0.44	
B02	Compliance classification		0.11		0.44	+0.33	
B21	Hallucination over-specification		0.28		0.61	+0.33	
B01	CCoP applicability scope		0.44		0.61	+0.17	
B12	Audit perspective alignment		0.33		0.50	+0.17	02 Flat where parametric memory covers it B07, B08, B09, B10 — general risk-and-compliance reasoning the base model already handles. Retrieval neither helps nor hurts.
B05	Control comprehension		0.22		0.28	+0.06	
B09	Risk identification & residual risk		0.72		0.78	+0.06	
B03	Conditional compliance reasoning		0.28		0.28	0.00	
B07	Gap identification quality		0.50		0.50	0.00	
B14	Remediation quality & feasibility		0.56		0.56	0.00	03 Two LLM-only failures rescued by RAG B02 and B18 score below the 15% pass threshold under Native (0.11 each). With retrieval, B02 lifts to 0.44 and B18 to 0.89.
B06	Intent understanding		0.28		0.22	-0.06	
B08	Risk-based prioritization		0.72		0.67	-0.05	
B10	Risk-justification coherence		0.56		0.50	-0.06	
B22	Waiver / exception reasoning		0.44		0.39	-0.05	
B24	Incident response guidance		0.22		0.17	-0.05	04 RAG hurts only at corpus boundaries B23 (multi-regulator MAS/PDPA/IM8), B24 (CII Regulations 2018 forms) reference material outside the seven-document corpus; B04 is well-covered by general training. Five regressions total — all corpus-coverage, not retrieval-mechanics.
B23	Multi-regulator coordination		0.33		0.17	-0.16	
B04	IT/OT classification boundary		0.78		0.44	-0.34	

WORKED EXAMPLE

Sample test showing *qualitative improvements* using RAG.

QUESTION • B13-001 "What evidence should a Risk Manager prepare to demonstrate board cybersecurity oversight?"		
NATIVE • LLM-ONLY		FAILED • 0.17
"From my training knowledge ... Clause 5.2.1 discusses the need for regular reporting to the board on risk management practices. Evidence should include Risk Assessment Reports, IRPs, Policies, Training Records — industry best practices suggest regular reporting and KPIs... it may not specify exact types of evidence. "		
RAG • HYBRID		PASSED • 0.61
"Cybersecurity reports to the board (Clause 3.2.1(d)); risk register updates with residual-risk monitoring (Clause 3.2.5); board meeting minutes reflecting cybersecurity discussions; cybersecurity exercise results (Clause 7.3.1); remediation plans (Clause 7.3.5)."		
METRIC / OBSERVATION	NATIVE (LLM-ONLY)	RAG (HYBRID)
VERDICT	Generic best-practices substitution; never identifies the specific board-oversight clauses (3.2.1(d), 3.2.5, 7.3.1, 7.3.5).	Cites all five board-oversight clauses with correct attribution.
HALLUCINATION — CLAUSE MISATTRIBUTION	" Clause 5.2.1 discusses regular reporting to the board... " — fabrication. Real 5.2.1 is about CII account privileges, not board reporting.	Not present.
HEDGING ON CLAUSE EXISTENCE	"... it may not specify exact types of evidence. However, industry best practices suggest..." — concedes inability to find specifics, then substitutes generic content.	Not present.
GENERIC-SUBSTITUTION PATTERN	Lists generic items (risk reports, IRP, policies, training records) any compliance framework would expect, without tying any to a CCoP clause.	Each item tied to a specific cited clause.
CITATION GROUNDING	" Clause 5.2.1 " — real clause ID, wrong substantive content (MISATTRIBUTED in the citation taxonomy).	Clauses 3.2.5, 3.2.1(d), 7.3.1, 7.3.5 — all real, all correctly described.
JUDGE DIMENSIONS	D1 0/3 D3 0/3 D6 0/3	D1 1/3 D3 2/3 D6 1/3
OUTCOME	FAILED • 0.17	PASSED • 0.61

— OBSERVATIONS

Where RAG helps · *where it does not.*

● Where & why RAG helps OBS 01 · 03 · 04

01

WHERE IT HELPS

RAG's contribution is dimensional, not uniform

+0.086 mean lift hides a sharply non-uniform pattern. Retrieval lifts D1 verdict (+0.09), D3 grounding (+0.15), D4 scope (+0.26), D6 citation (+0.09) — the dimensions that depend on knowing the right CCoP clause and what it says. RAG is a tool for correctness against external authority, not for reasoning quality.

03

WHY IT HELPS

Helps most where the topic is known but the clause is not

The four > +0.30 wins (B18, B13, B02, B21) share a property: the topic is well-represented in training, but the specific CCoP clause text is not. With retrieval, the same model with the same prompt produces the right answer — burden shifts from recall to read.

04

WHERE IT HELPS

Native LLM-only is structurally hazardous

Two of 18 cases (B02, B18) fell below the 15% pass threshold under Native — with confidently-asserted specifics that read as accurate citations. Confidently wrong is qualitatively worse than acknowledged uncertainty in a compliance domain. RAG removes that failure mode.

● Where it fails · what's missing OBS 02 · 05

02

WHAT'S MISSING

Reasoning is solid; the gap is fact recall

D2 stability (0.593 → 0.611) is the strongest signal that the base model reasons adequately. What it lacks is reliable parametric memory of which CCoP clause says what. Future fine-tuning is best targeted at narrow factual recall, not reasoning capability.

05

WHERE IT FAILS

Underperforms only at corpus boundaries

Five regressions (B04, B23, B24, marginal B05 / B22) trace to corpus-coverage failures, not retrieval mechanics. B23 needs MAS / PDPA / IM8; B24 needs CII Regulations 2018 forms — material outside the seven-document corpus. The highest-leverage next step is corpus expansion.

NEXT STEPS

Six work items, *one sequence.*

01

Ablation study using GraphRAG

OPTIONAL • EXPLORATORY

Index documents as a knowledge graph rather than flat chunks. For a regulatory corpus with cross-clause references (Cybersecurity Act §11(7); Section 7 → 8 BCP), graph-structured retrieval may surface multi-hop dependencies the flat-chunk index misses.

02

Domain-expert ground-truth confirmation

LOW-RISK DEPENDENCY

Auditor + verifier agent pair has produced 442 verifier-approved corrections. The remaining task is human-expert sign-off across the 18 active benchmarks and the 435-case corpus — confirming, not reconstructing.

03

Re-run the full evaluation suite

AFTER ITEM 02

Re-execute the 18-case stratified sample plus the full 435-case corpus under both llm-only and hybrid modes once expert validation closes. Dimensional pattern and benchmark-level deltas should hold.

04

Close corpus gaps where RAG underperforms

CORPUS EXPANSION

Five RAG-negative benchmarks (B04, B23, B24, marginal B05 / B22) are corpus-coverage failures. Enumerate per benchmark — CII Regulations 2018, MAS TRM, PDPA, IM8, GovTech directives — and ingest into the retrieval index.

05

Generate fine-tuning dataset from validated GT

TERM 3 PREP

Use audit-corrected and expert-validated GT to derive a CCoP-aligned training set targeting the factual-recall gap. Emphasise clause-content alignment, not reasoning chains — those are already adequate.

06

Research RAFT vs vanilla QLoRA fine-tuning

OPEN QUESTION

RAFT (Zhang et al., 2024) trains the model on retrieved-context-plus-distractors, teaching grounding rather than memorisation. Given the production system uses retrieval at inference, is RAFT a better paradigm than vanilla QLoRA?

— TERM 3 • A PREVIEW

Retrieval-Augmented Fine-Tuning — *RAFT*.

RAFT trains the model on retrieved-context-plus-distractors so it learns to ground responses in retrieved evidence and ignore noise — combining the dynamic knowledge of RAG with the domain skill that fine-tuning provides.

DIMENSION	RETRIEVAL ONLY Standard RAG	STATIC MEMORISE Fine-Tuning	RETRIEVAL-AWARE RAFT
KNOWLEDGE SOURCE	External database (dynamic)	Model weights (static)	External database + extraction skill
UPDATE EFFORT	Low — just update the database.	High — requires retraining / fine-tuning.	Low — skill is learned; data stays external.
HALLUCINATION RISK	Medium — can get confused by noise.	High — confidently makes up facts.	Low — trained to ignore noise and cite text.
NOISE RESISTANCE	Weak — struggles with irrelevant "distractors".	N/A — relies on internal memory.	Strong — specifically trained to filter out junk.
AUDITABILITY	High — provides source documents.	Low — "black box" internal weights.	Very high — cites specific clauses via CoT.
TRAINING GOAL	None (in-context learning).	Fact / style memorisation.	Retrieval-aware reasoning.

¹ Zhang, T. et al. (2024). RAFT: Adapting Language Model to Domain Specific RAG. arXiv:2403.10131 — <https://arxiv.org/abs/2403.10131>

Thank *you.*

Questions, comments, and pushback all welcome.

